Exp:3
30/07/2025

Learning And assignment of IP Address Manually To Computers.

AIM:

To understand the IP addressing Scheme and manually assign static IP address to computer on a network.

Theory:

An IP (Internet Protocol) adress. is a unique numerical lable assignment to each device connected to a computer network IPv4 address are 32 bit number written in dotted decimal notation.

Subnet mark determines the network and host positions. Manual (Static) IP assignment is used for servers and network devices required a permanet address DHCP assigns IPs dynamically

Procedure:

In windows: Open network and Sharing center → Change adapter settings → Right click

local area Connection → Properties → IPv4 → Set IP manually

→ Enter IP address: 192.168.1.10, Subnet mask: 255.255.255.0, Default gateway: 192.168.1.1.

→ Enter DNS server: 8.8.8.8 (google DNS) click Ok.

→ In linux: use Command 'sudo ip addr add 192.168.10/24 dev eth0',

→ To get gateway in linux: "sudo ip route add default via 192.168.1.1".

→ for Persistent Configuration in linux, edit /etc/network/interfaces or use nmcli.
→ verify the configuration using 'ipconfig' or 'ifa'
→ Ping the default gateway to confirm network connectivity.
→ Ping another computer on the same subnet to verify communication.

Result:

Static IP address were successfully assigned to computers connectivity was verified using Ping commands b/w hosts on the same subnet.